



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Department of Computer Science

CSC 3215: Web Technologies

Semester: Summer 2025-26

Group No:

Section: Q

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Project Title

CareerLink: Online Job Portal

Short Description

CareerLink is a web-based online job portal that connects job seekers with employers through a simple and organized recruitment system. Job seekers can search and filter available jobs, view job details, apply for suitable positions by submitting their CVs and track their application status. Employers can post and manage job circulars, view applicants, download CVs and review applications. An administrator manages users, job posts and platform statistics.

Introduction

Finding suitable employment opportunities and managing recruitment can be challenging when job information and applications are handled through different sources. CareerLink aims to provide a centralized web-based platform where job seekers can easily discover and apply for suitable jobs, while employers can efficiently post vacancies and manage applications. The system provides separate functionalities for Job Seekers, Employers and Administrators. Job Seekers can search and filter jobs, view job details, submit applications with CVs and track their application status. Employers can create and manage job postings, view applicants, download CVs and update application status. Administrators can manage users and job postings and monitor overall platform statistics.

Background Study

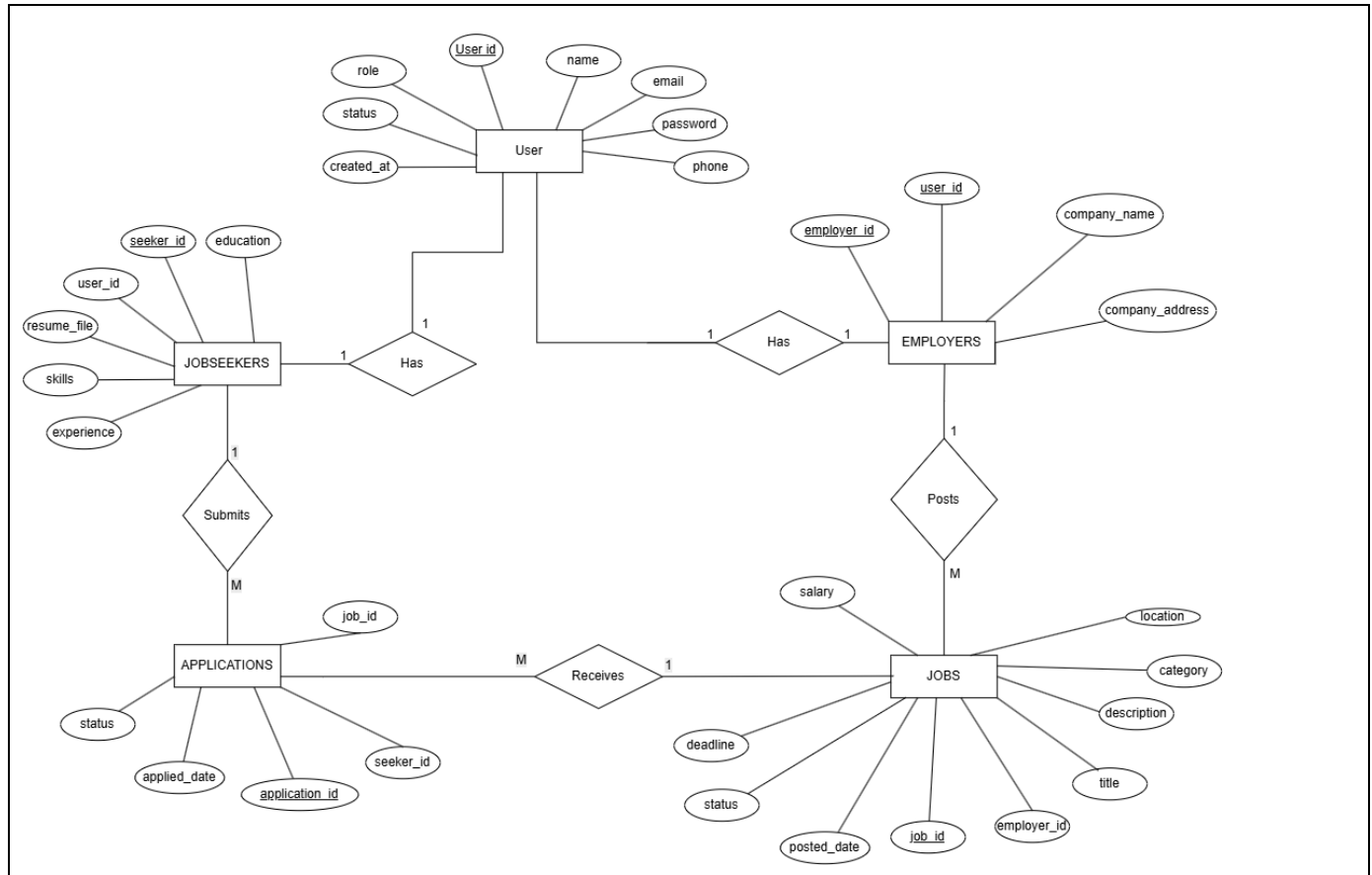
Nowadays, job searching and recruitment are increasingly performed through online platforms. Traditional recruitment processes can be time-consuming because job seekers often search through different sources and employers need to manage job posts, applications and resumes separately.

CareerLink is designed as a centralized online job portal to simplify this process. Job seekers can search and filter jobs, apply with their CVs and track application status. Employers can post and manage job circulars, view

applicants, download CVs and review applications. Administrators can manage users and job posts while monitoring basic system statistics.

The project focuses on essential recruitment features to keep the system simple and manageable. Advanced features such as automatic job recommendations, real-time messaging, interview scheduling and resume analysis are kept outside the current scope and may be added in future versions.

ER Diagram



Normalization:

❖ HAS — Users & JobSeekers

UNF: HAS_SEEKER(user_id, name, email, password, phone, role, status, created_at, seeker_id, resume_file, skills, education, experience)

1NF: No repeating/multivalued attributes → relation is in 1NF.

2NF: Partial dependencies exist:

user_id → name, email, password, phone, role, status, created_at

seeker_id → resume_file, skills, education, experience, user_id.

Decompose into:

- USERS (user_id, name, email, password, phone, role, status, created_at)
- JOBSEEKERS (seeker_id, resume_file, skills, education, experience, user_id)

3NF: No transitive dependency in the decomposed relations → both are in 3NF.

❖ HAS — Users & Employers

UNF:

HAS_EMPLOYER (user_id, name, email, password, phone, role, status, created_at, employer_id, company_name, company_address)

1NF: No repeating/multivalued attributes → relation is in 1NF.

2NF: Partial dependencies exist:

user_id → name, email, password, phone, role, status, created_at

employer_id → company_name, company_address, user_id.

Decompose into:

i. USERS (**user_id**, name, email, password, phone, role, status, created_at)

ii. EMPLOYERS (**employer_id**, company_name, company_address, user_id)

3NF: No transitive dependency in the decomposed relations → both are in 3NF.

❖ POSTS — Employers & Jobs

UNF:

POSTS (employer_id, company_name, company_address, job_id, title, description, category, location, salary, deadline, status, posted_date)

1NF: No repeating/multivalued attributes → relation is in 1NF.

2NF: Partial dependencies exist:

employer_id → company_name, company_address

job_id → title, description, category, location, salary, deadline, status, posted_date, employer_id.

Decompose into:

i. EMPLOYERS (**employer_id**, company_name, company_address, user_id)

ii. JOBS (**job_id**, title, description, category, location, salary, deadline, status, posted_date, employer_id)

3NF: Employer information is separated from JOBS, so no transitive dependency remains → both are in 3NF.

❖ SUBMITS — JobSeekers & Applications

UNF:

SUBMITS (seeker_id, resume_file, skills, education, experience, application_id, applied_date, status)

1NF: No repeating/multivalued attributes → relation is in 1NF.

2NF: Partial dependencies exist:

seeker_id → resume_file, skills, education, experience

application_id → seeker_id, applied_date, status.

Decompose into:

i. JOBSEEKERS (**seeker_id**, resume_file, skills, education, experience, user_id)

ii. APPLICATIONS (**application_id**, seeker_id, applied_date, status)

3NF: No transitive dependency in the decomposed relations → both are in 3NF.

❖ RECEIVES — Jobs & Applications

UNF:

RECEIVES (job_id, title, description, category, location, salary, deadline, status, posted_date, application_id, applied_date, application_status)

1NF: No repeating/multivalued attributes → relation is in 1NF.

2NF: Partial dependencies exist:

job_id → title, description, category, location, salary, deadline, status, posted_date

application_id → job_id, applied_date, application_status.

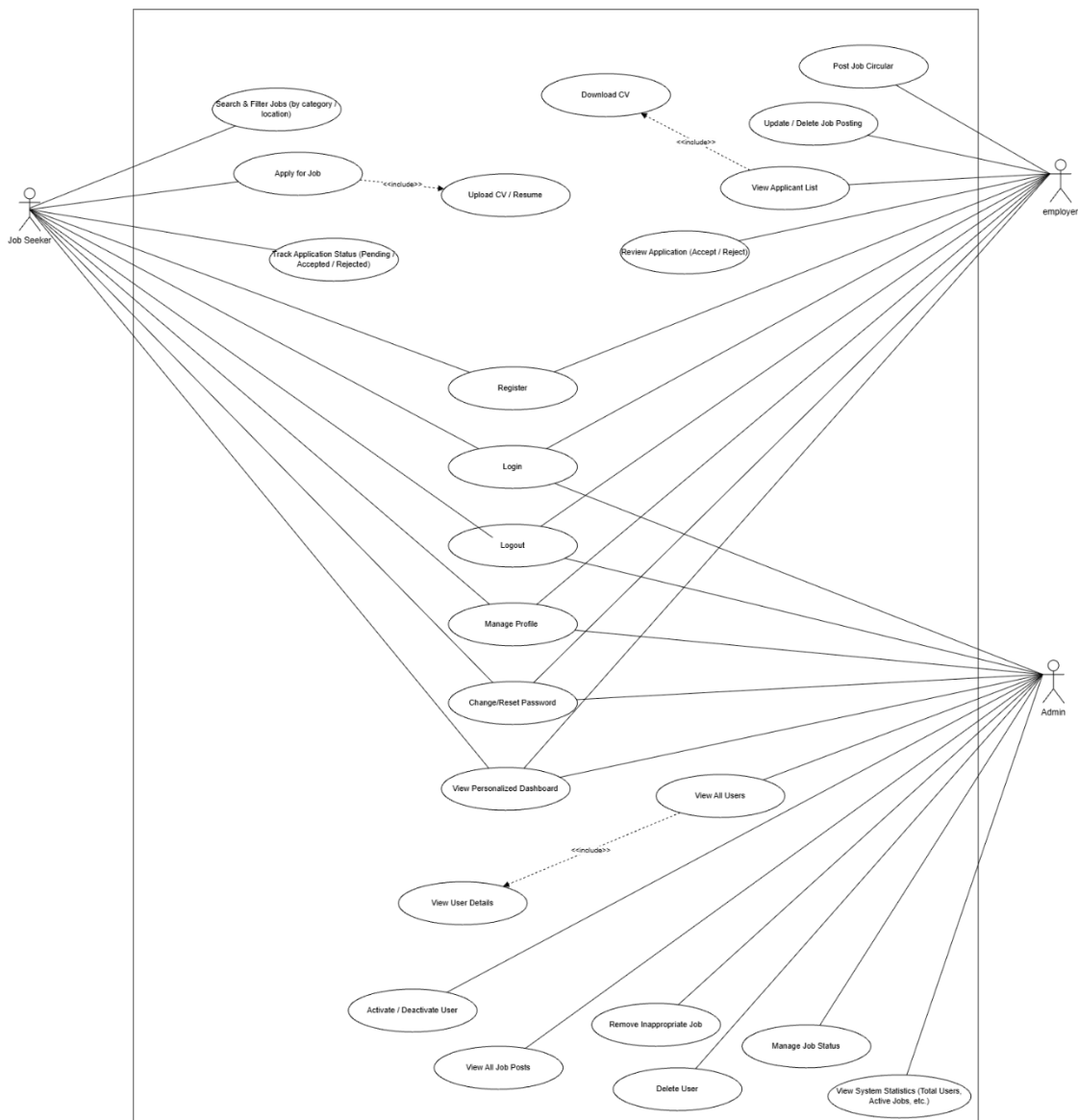
Decompose into:

- i. JOBS (job_id, title, description, category, location, salary, deadline, status, posted_date, employer_id)
 - ii. APPLICATIONS (application_id, job_id, applied_date, status)
- 3NF:** No transitive dependency in the decomposed relations → both are in 3NF.

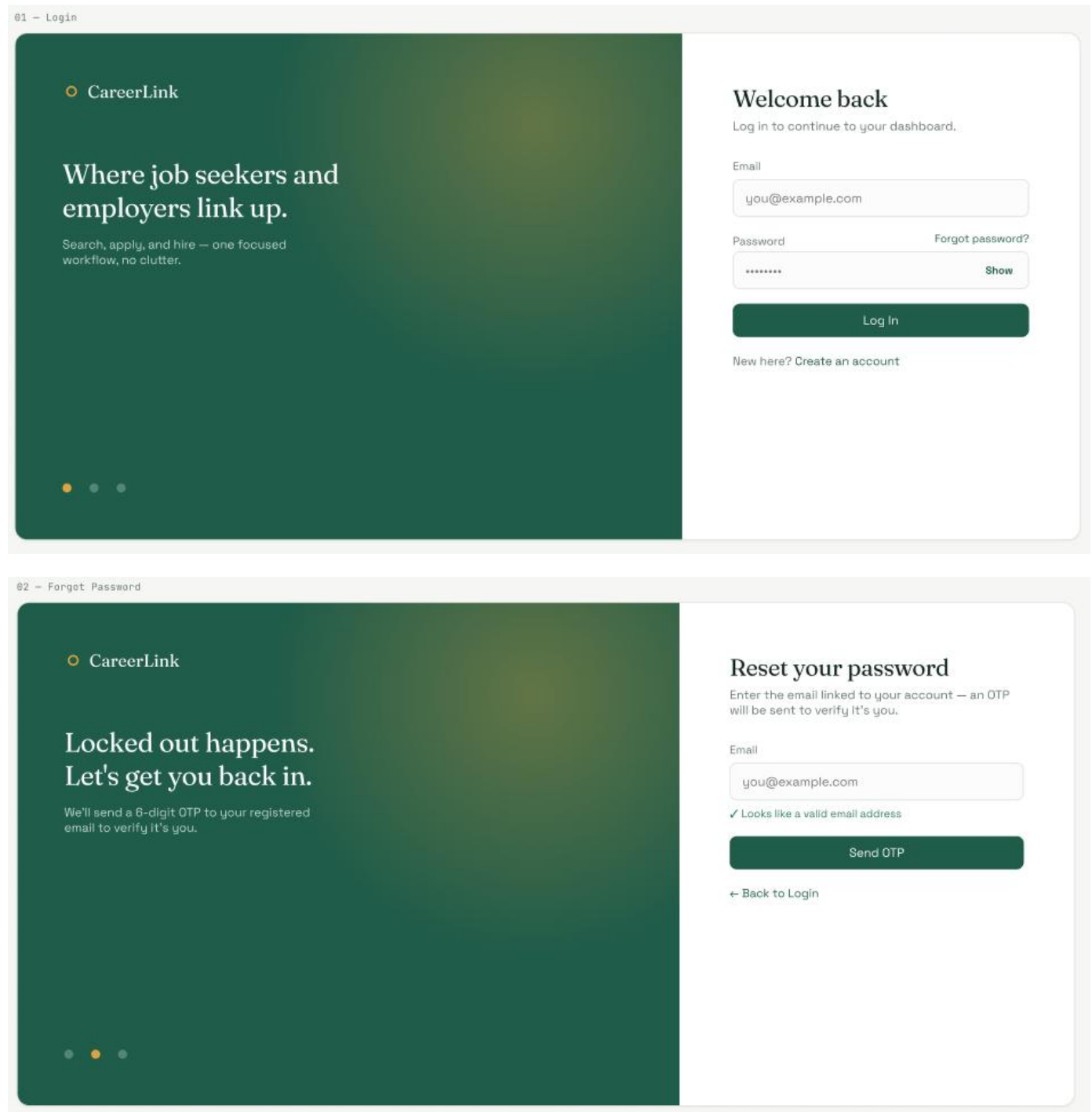
❖ Final Normalized Tables

1. USERS(user_id PK, name, email, password, phone, role, status, created_at)
2. JOBSEEKERS(seeker_id PK, user_id FK, resume_file, skills, education, experience)
3. EMPLOYERS(employer_id PK, user_id FK, company_name, company_address)
4. JOBS(job_id PK, employer_id FK, title, description, category, location, salary, deadline, status, posted_date)
5. APPLICATIONS(application_id PK, job_id FK, seeker_id FK, applied_date, status)

Use Case Diagram



UI (Figma/Sketch)



 CareerLink

Almost there — verify your code.

Enter the 6-digit OTP we sent to your registered email to continue.



Verify OTP

Code sent to ar****@example.com. Enter it below to continue.

6-Digit OTP

⌚ Expires in 05:00

Code expires 5 minutes after it's sent

3 incorrect attempts remaining before the code is locked

Verify OTP

Didn't get the code? Resend OTP

← Back to Login

 CareerLink

One last step.

Choose a new password to secure your account.



Create new password

Your OTP has been verified. Set a new password below.

New Password

Show

Confirm Password

Show

✓ Minimum 8 characters · passwords must match

Reset Password

← Back to Login

CareerLink

You're all set.

Your account is secured with your new password.



Password reset successful

You can now log in to CareerLink using your new password.

[Back to Login](#)

CareerLink

Create your account in under a minute.

Pick your role — the rest of the platform adapts to it.

Create account

Register as a job seeker or an employer.

☒ Job Seeker

☐ Employer

Full Name

Arafat Hossain

Email

you@example.com

Phone

01XXXXXXXX

Password

[Show](#)

[Register](#)

Already have an account? [Log in](#)

DashboardBrowse JobsMy ApplicationsProfileAR

Welcome back, Arafat

Here's where your applications stand.

12
Total Applications

5
Pending

4
Accepted

3
Rejected

Recent Applications

[View all](#)

JOB TITLE	COMPANY	APPLIED	STATUS
Junior PHP Developer	Nexbridge Ltd.	Aug 12	Accepted
Frontend Engineer	Studio Loom	Aug 15	Pending
Network Support Intern	ConnectIT	Aug 09	Rejected

DashboardBrowse JobsMy ApplicationsProfileAR

All CategoriesAll LocationsSearch

Junior PHP Developer

Web DevelopmentNexbridge Ltd. Dhaka ₳30,000-40,000

View Details

Frontend Engineer (React)

Web DevelopmentStudio Loom Remote ₳45,000-60,000

View Details

Network Support Intern

NetworkingConnectIT Chattogram ₳15,000

View Details

DashboardBrowse JobsMy ApplicationsProfileAR

[← Back to Jobs](#) Web Development

Junior PHP Developer

Nexbridge Ltd. · Dhaka · Posted Aug 10

Description

We're looking for a junior PHP developer comfortable with MVC-style codebases to help maintain and extend our internal tools. You'll work closely with two senior engineers.

Requirements

PHP, MySQL, basic Git workflow, willingness to learn our internal MVC framework.

Apply for this job

Resume / CV

Choose file — resume.pdf

Submit Application

Salary: ₳30,000-40,000
Deadline: Sep 05, 2026

CareerLink

DashboardBrowse JobsMy ApplicationsProfile

AR

[← Back to Dashboard](#)

My Applications

JOB TITLE	COMPANY	APPLIED DATE	STATUS
Junior PHP Developer	Nexbridge Ltd.	Aug 12, 2026	Accepted
Frontend Engineer	Studio Loom	Aug 15, 2026	Pending
Network Support Intern	ConnectIT	Aug 09, 2026	Rejected
QA Tester	PixelWorks	Aug 18, 2026	Pending

CareerLink

DashboardBrowse JobsMy ApplicationsProfile

AR

[← Back to Dashboard](#)

Full Name

Arafat Hossain

Email

arafat@example.com

Phone

01XXXXXXXXX

Education

BSc in Computer Science, AIUB (2023–present)

Experience

6-month intern, local web agency

Skills

PHP, MySQL, JavaScript, HTML, CSS

Save Changes

Resume

resume_arafat.pdf

Replace File

Danger Zone

Permanently delete your account and all application history.

Delete Account

 CareerLink

DashboardMy JobsPost JobProfile

NX

Nexbridge Ltd.

Your hiring activity at a glance.

6
Jobs Posted

4
Active Jobs

37
Total Applicants

9
Pending Review

Recently Posted Jobs

+ Post New Job

Junior PHP Developer

Active12 applicants

Manage

DevOps Assistant

Closed25 applicants

Manage

 CareerLink

DashboardMy JobsPost JobProfile

NX

Post a New Job Circular

Job Title

e.g. Junior PHP Developer

Description

Describe the role...

Requirements

Skills, experience needed...

Category

Web Development

Location

Dhaka

Salary Range

৳30,000-40,000

Application Deadline

Sep 05, 2026

Post Job

My Job Postings

TITLE	LOCATION	DEADLINE	STATUS	APPLICANTS	
Junior PHP Developer	Dhaka	Sep 05, 2026	Active	12	<button>Edit</button> <button>Delete</button>
DevOps Assistant	Remote	Aug 01, 2026	Closed	25	<button>Edit</button> <button>Delete</button>
UI/UX Intern	Dhaka	Sep 20, 2026	Active	4	<button>Edit</button> <button>Delete</button>

[← Back to My Jobs](#)

My Jobs / Junior PHP Developer

Applicants (12)

APPLICANT	APPLIED	RESUME	STATUS	ACTION
Arafat Hossain	Aug 12	Download CV	Accepted	<button>Accepted</button>
Nusrat Jahan	Aug 13	Download CV	Pending	<button>Pending</button>
Rifat Kabir	Aug 14	Download CV	Rejected	<button>Rejected</button>

[← Back to Dashboard](#)

Full Name

Nexbridge HR

Email

hr@nexbridge.com

Phone

02XXXXXXXXX

Company Name

Nexbridge Ltd.

Company Address

House 12, Road 5, Dhaka

Save
Changes

Danger Zone

Permanently delete this employer account and all posted jobs.

Delete Account

CareerLink

Admin

Dashboard

Manage Users

Manage Jobs

Statistics

AD

System Overview

248
Total Users

176
Job Seekers

72
Employers

54
Active Jobs

Recently Flagged

Data Entry — Work From Home
Posted by QuickCash Ltd. · reported 3 times

Remove Job

CareerLink

Admin

Dashboard

Manage Users

Manage Jobs

Statistics

AD

Search by name or email...

All Roles

NAME	EMAIL	ROLE	STATUS	ACTION
Arafat Hossain	arafat@example.com	Job Seeker	Active	View Details Deactivate Delete
Nexbridge Ltd.	hr@nexbridge.com	Employer	Active	View Details Deactivate Delete
QuickCash Ltd.	contact@quickcash.com	Employer	Inactive	View Details Activate Delete

CareerLink

Admin

Dashboard

Manage Users

Manage Jobs

Statistics

AD

← Back to Manage Users

Name

Arafat Hossain

Email

arafat@example.com

Phone

01XXXXXXXX

Role

Job Seeker

Account Status

Active

Registration Date

Jul 02, 2026

Admin Actions

Deactivate User

Delete User

20 - Admin / Manage Jobs

CareerLink Admin	Dashboard	Manage Users	Manage Jobs	Statistics	AD
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← Back to Dashboard

JOB TITLE	EMPLOYER	STATUS	ACTION
Junior PHP Developer	Nexbridge Ltd.	Active	Deactivate
Data Entry — Work From Home	QuickCash Ltd.	Removed	Restore
UI/UX Intern	Nexbridge Ltd.	Active	Deactivate

21 - Admin / System Statistics

CareerLink Admin	Dashboard	Manage Users	Manage Jobs	Statistics	AD
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Platform Statistics

248 Total Users	89 Total Jobs Posted	54 Active Jobs	431 Total Applications
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User Breakdown

176 Job Seekers · 71 Employers · 1 Admin

Types of Users: 1. Job-Seeker 2. Employer 3. Admin

Common Features (Available to All Users)

- **Authentication**
 1. Login to the system
 2. Logout from the system
 3. User registration
- **Account Management**
 1. Change or reset password
 2. Manage profile information (view, edit, delete)
- **Dashboard**
 1. Access a personalized dashboard after login

Specific Features by User Type

- **Features of Job Seeker**
 1. Search and filter job opportunities
 2. Apply for jobs and upload CV
 3. Track application status
- **Features of Employer**
 1. Posting and manage job circulars
 2. View applicants and download CVs
 3. Review applications and update applicant status

- **Features of Admin**

1. Manage registered users
2. Manage and monitor job postings
3. View platform statistics